Telco Customer Churn Prediction

Binary classification model to rank customers by churn risk

Dataset	1A_telco_customer_churn.csv (PA-1A)
Rows x Columns	7043 x 26
Target	Churn (Yes/No, ~26.5% positive)
Primary metric	PR-AUC
Random state	42

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The retention team must prioritize which customers to contact first

A telecom company loses customers every month. With a limited retention budget, the team needs to know which customers to contact first.

Decision supported: Rank customers by churn risk; prioritize outreach to those most likely to churn.

Target: Churn (1 = Yes, 0 = No)

Success metrics: PR-AUC (primary), ROC-AUC, precision, recall, F1, confusion matrix. Business: expected saved revenue and lift at top-20%.

Why PR-AUC: Classes are imbalanced (~26.5% churn). Recall matters (missed churners = lost revenue); precision matters (wrong contacts waste budget).

Data: 7,043 customers, 21 features, 26.5% churn rate

Dataset size: 7043 rows x 26 columns

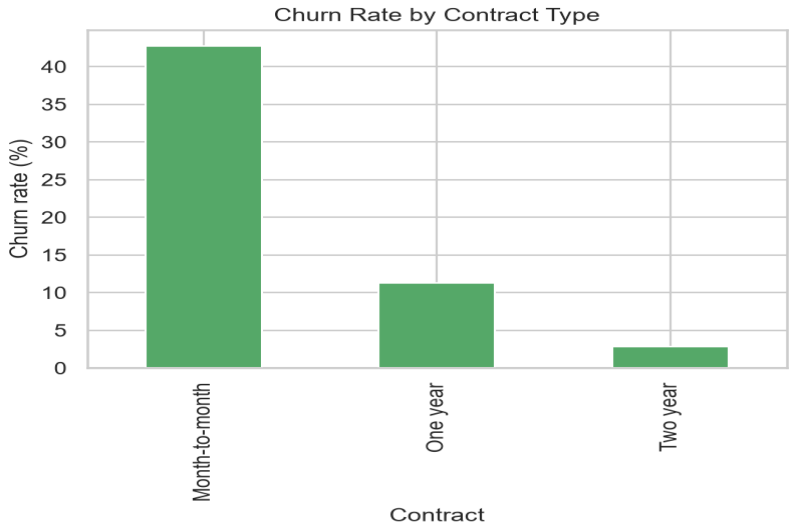
Class balance: 73.5% No churn, 26.5% Yes churn

Data quality:

- Missing values: 11 blank strings in TotalCharges (coerced to NaN, filled with 0)
- Data types: TotalCharges read as string, coerced to numeric
- Duplicates: 0 duplicate rows; customerID is unique
- Outliers: Checked via IQR on tenure, MonthlyCharges, TotalCharges; kept as plausible values

TotalCharges blank strings occur mostly for tenure=0 customers (no accumulated charges yet).

Month-to-month customers churn at 42.7%, vs 2.8% on two-year contracts

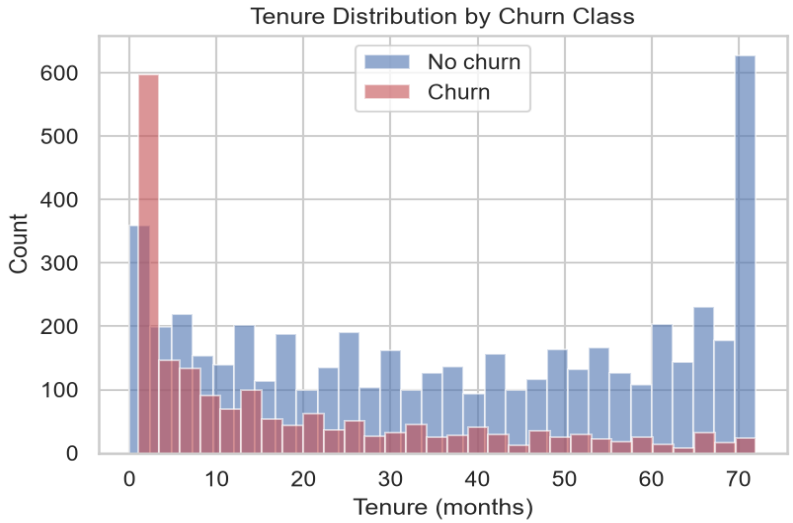


Finding: Contract type is the strongest churn predictor.

- Month-to-month: 42.7% churn
- One-year: 11.3% churn
- Two-year: 2.8% churn

Implication: Longer contracts act as switching-cost barriers. Prioritize month-to-month customers in the top-risk segment.

New customers churn more: mean tenure 18.0 vs 37.6 months

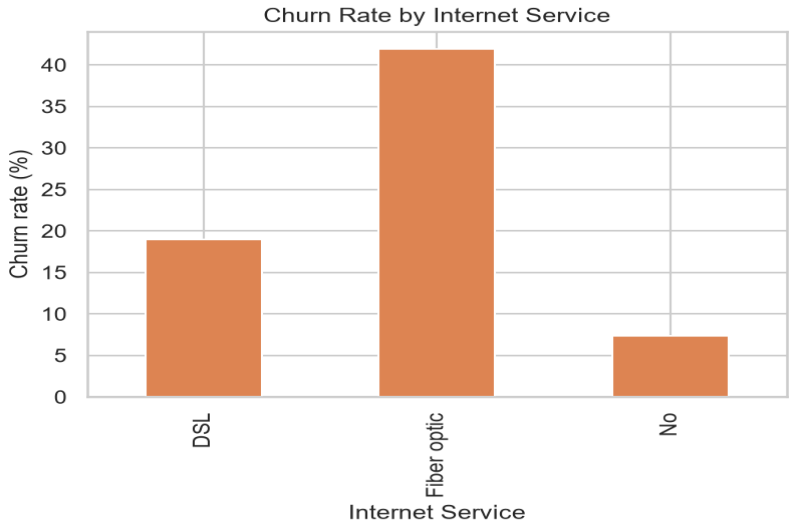


Finding: Tenure is strongly associated with churn risk.

- Mean tenure, non-churners: 37.6 months
- Mean tenure, churners: 18.0 months

Implication: New customers (tenure <= 12 months) are most vulnerable. An onboarding/check-in program is warranted.

Fiber optic customers churn at 41.9%, vs 7.4% with no internet



Finding: Internet service type affects churn risk.

- Fiber optic: 41.9% churn
- DSL: 19.0% churn
- No internet: 7.4% churn

Implication: Service-quality follow-ups for fiber customers in the high-risk segment.

Approach: pipeline preprocessing, 5 engineered features, stratified 60/20/20 split

Preprocessing: Numeric: median imputation + StandardScaler. Categorical: OneHotEncoder(handle_unknown=ignore). All inside a ColumnTransformer within a Pipeline (no leakage).

Engineered features (5):

- avg_monthly_spend - removed (highly collinear with MonthlyCharges, r=0.956)
- num_addon_services - count of Yes among 6 add-on services
- is_new_customer - 1 if tenure <= 12
- has_long_contract - 1 if not month-to-month
- charge_tenure_ratio - MonthlyCharges / (tenure + 1)

Validation: Stratified split: 60% train, 20% validation, 20% test. Test set used once only, in final evaluation.

Logistic Regression leads on PR-AUC and interpretability

Model comparison on validation set (test set reserved for final evaluation):

Model	ROC-AUC	PR-AUC	Precision	Recall	F1	Interpretability
Logistic Regression	0.840	0.654	0.541	0.735	0.624	-
Random Forest	0.826	0.618	0.566	0.666	0.612	-
Baseline (Dummy)	0.500	0.265	0.000	0.000	0.000	-

Tuning (Random Forest): Default PR-AUC 0.6175 -> tuned PR-AUC 0.6348 (+0.0173). Best params: {'clf__max_depth': 14, 'clf__min_samples_leaf': 1, 'clf__n_estimators': 400}

Final model: Logistic Regression - highest validation PR-AUC and superior interpretability (coefficients show direction of effect).

Logistic Regression: test ROC-AUC 0.846, PR-AUC 0.663

Final model (retrained on train + validation, evaluated once on test):

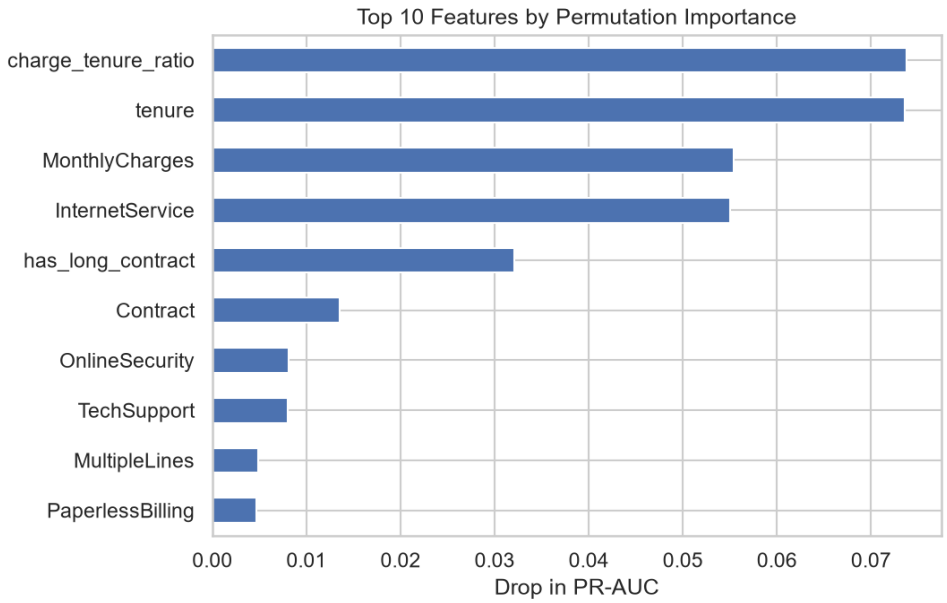
- ROC-AUC: 0.846 | PR-AUC: 0.663
- Confusion matrix: TN=796, FP=239, FN=92, TP=282

Random Forest (comparison): ROC-AUC 0.836, PR-AUC 0.627

Threshold: Top-20% (80th percentile), not default 0.5.

- Customers flagged: 282 of 1409
- Precision: 0.667 | Recall: 0.503
- Lift at top-20%: 1.89x

charge_tenure_ratio and Contract are the top predictors



Top features (permutation importance, drop in PR-AUC):

- charge_tenure_ratio: 0.0737
- tenure: 0.0736
- MonthlyCharges: 0.0554
- InternetService: 0.0550

Direction matches expectations: Contract, tenure, has_long_contract reduce churn; charge_tenure_ratio increases it. is_new_customer ranks below tenure (overlapping information). num_addon_services is not in top 10.

Three retention recommendations with estimated net impact

Rec 1 (High): Offer discounted migration to 1-year contracts for flagged month-to-month customers. Net impact: 62,619 per period

Assumptions: 40% migration success, 15 discount cost per customer. 187 churners captured among 280 flagged month-to-month.

Rec 2 (High): 90-day onboarding and check-in program for flagged new customers (tenure <= 12). Net impact: 41,660 per period

Assumptions: 35% retention success, 20 per customer. 147 churners captured among 215 flagged new customers.

Rec 3 (Medium): Service-quality follow-up for flagged fiber optic customers. Net impact: 28,918 per period

Assumptions: 25% retention success, 30 per customer. 162 churners captured among 242 flagged fiber customers.

Whole program: Net impact 43,332 (top-20% of 1409 test customers, 188 churners captured, lift 1.89x).
Sensitivity: 21,666 (15% retention) to 64,998 (45% retention).

Limitations and next steps

Limitations:

- All monetary values in dataset monetary units (currency unspecified in data)
- Business assumptions (30% retention success, 25 cost per contact) need validation
- No external data used; results limited to this dataset
- Deep learning, NLP, computer vision, and deployment out of scope

Next steps:

- Validate assumptions with the business team
- A/B test the retention program on a pilot segment
- Consider cost-sensitive threshold tuning based on actual contact costs
- Monitor model performance over time with new data

AI Usage Disclosure (Appendix)

Tool: Hermes Agent by Nous Research

Parts assisted by AI:

- Project scaffolding and environment setup (uv, virtual environment, dependencies)
- Notebook code generation based on the build guide
- Data exploration, modeling, and business impact calculation templates

Independent work:

- Choice of primary metric (PR-AUC), threshold rule (top-20%), class imbalance approach
- Feature selection decisions, final model choice (Logistic Regression)
- Interpretation of results, recommendation selection, sensitivity parameters

Verification: All code executed top-to-bottom with no errors. All numbers reproducible by running code cells.